

WASP-50b

R-0899

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-50_230918 · created 2026-07-29T03:15:56+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460205.856 +/- 0.094 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.142 +/- 0.049
Transit depth (Rp/Rs)^2	2.02 +/- 1.39 [%]
Orbital Inclination (inc)	83.0 +/- 2.87
Airmass coefficient 1 (a1)	0.926 +/- 0.012
Airmass coefficient 2 (a2)	0.036 +/- 0.012
Scatter in the residuals of the lightcurve fit is	1.29 %
Best Comparison Star	#1 - [280, 157]
Optimal Aperture	4.12
Optimal Annulus	18.92
Transit Duration (day)	0.018 +/- 0.025

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.142 ± 0.049 vs 0.1404 (deviation 0.03σ)
Duration fit vs published	0.018 d vs 0.0754 d (deviation 2.3σ)
Mid-time O–C	-18.0 min
Depth SNR	2.9
Residual scatter	1.29 %

Light curve

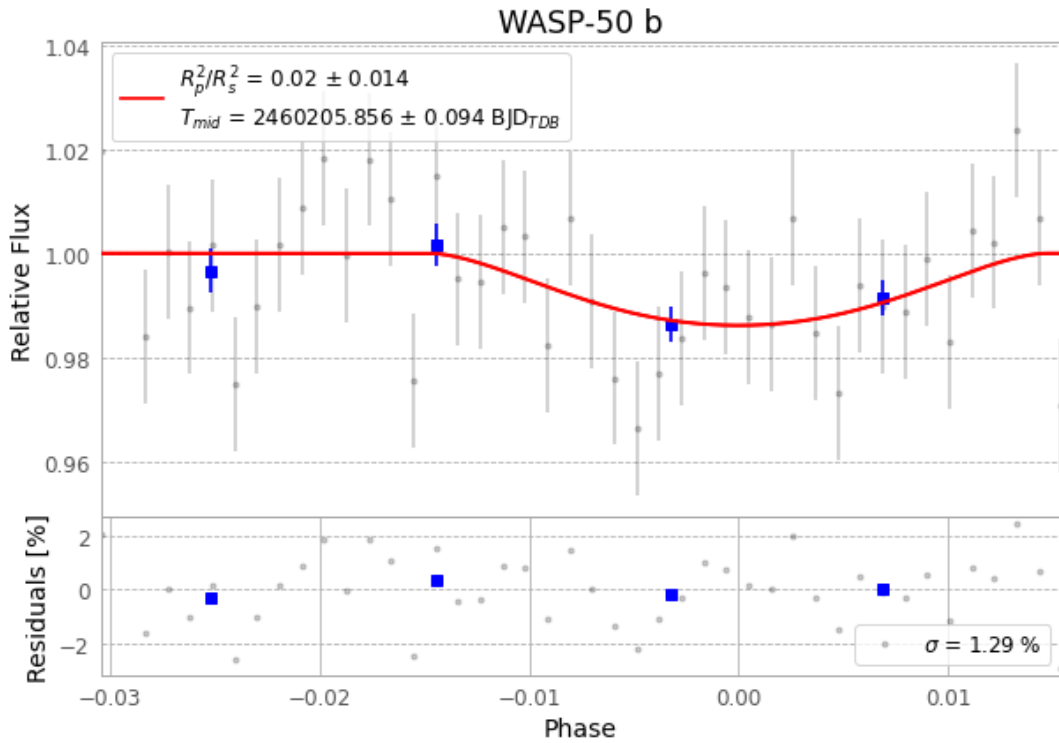


Figure 1 — FinalLightCurve_WASP-50 b_2023-09-17.png (SHA-256 d8f6c72cac9d1825...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [337, 220], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 8f98bcef9ee0871e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

45 FITS frames (30 MB), WASP-50230918065715.FITS → WASP-50230918090915.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-50-230918.log (f3bfcfb4a495...), FinalLightCurve_WASP-50 b_2023-09-17.png (d8f6c72cac9d...), FinalLightCurve_WASP-50 b_2023-09-17.csv (6d45c1ecca14...)

Compute resources

Wall time 75.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 03:15Z.